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When a Derivative Becomes a Distribution:
A quiet decade of bias in coercivity unmixing

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Abstract

Magnetic coercivity unmixing has become a standard tool in rock and environmental magnetism. Over the past two decades, a multitude of studies has relied on coercivity spectra to interpret magnetic mineral assemblages in rocks and sediments, soils and other environmental archives. Like many widely used analytical techniques, however, the method rests on a series of mathematical assumptions that are rarely examined once the software becomes familiar. In this article I explore one such assumption embedded in the popular MAX UnMix program and show how a seemingly innocuous mathematical choice, the absolute value of a derivative, can systematically transform measurement noise into apparent magnetic components.

1. A useful idea

Coercivity unmixing has become one of the most widely used analytical tools in rock and environmental magnetism. The approach relies on the observation that the shape of an IRM acquisition or backfield curve for a population of grains of a single magnetic mineral can often be approximated by a cumulative log-Gaussian distribution.

Kruiver et al. (2001) formalized this concept through cumulative log-Gaussian (CLG) analysis, in which the same IRM dataset is examined in three complementary representations: the linear

acquisition plot (LAP); the gradient acquisition plot (GAP), and the standardized acquisition plot (SAP).

Together these representations allow an IRM curve to be decomposed into a mixture of coercivity distributions characterized by mean coercivity and dispersion parameters.

Importantly, Kruiver et al. emphasized that the gradient representation is inherently sensitive to noise because it relies on a numerical derivative of the data. For this reason, the LAP, GAP, and SAP representations were intended to be interpreted together, providing mutual constraints during forward modeling.

2. The arrival of MAX UnMix

In 2016, MAX UnMix (Maxbauer et al., 2016) was introduced as a web-based tool designed to make coercivity unmixing more accessible to the community.

The program implemented a mixture model of skew-normal probability density functions and included Monte-Carlo resampling to estimate uncertainties in model parameters.

The statistical model underlying MAX UnMix defines the observed coercivity distribution as

$$C = \left| \frac{dM}{d\log B} \right| \tag{1}$$

that is, the absolute value of the derivative of magnetization with respect to the logarithmic field.

At first glance this definition appears harmless. Coercivity spectra have long been plotted as positive distributions. However, once measurement noise is considered, the statistical implications become important.

3. When noise becomes signal

At sufficiently high and low fields, the true coercivity distribution approaches zero:

$$\frac{dM}{d\log B} \to 0 \tag{2}$$

In these regions the derivative of the data is dominated by measurement noise. If the measurement noise is symmetric around zero, the derivative fluctuates around zero and its expected value is

$$E\left[\frac{dM}{d\log B}\right] = 0 \tag{3}$$

Taking the absolute value fundamentally changes the statistics. For any symmetric random variable X,

$$E(|X|) > 0 \tag{4}$$

Consequently, even when the true derivative is zero, the expected value of the computed coercivity distribution becomes positive.

In practical terms, noise in the derivative is transformed into a strictly positive signal. The result is a systematic upward bias in the tails of the coercivity spectrum that increases with noise level.

4. A synthetic test

To illustrate this effect, a synthetic IRM acquisition curve consisting of a single coercivity component was generated with a peak at 50 mT (dispersion parameter sigma= 0.25, skewness= 0).

The dataset contains 100 logarithmically spaced acquisition points. Noise levels of 0, 0.5, 1, 2, 5, and 10% were added, and the resulting datasets were processed using MAX UnMix to generate gradient curves.

Clean dataset

In the absence of noise, both the cumulative curve and its derivative produce the expected single Gaussian distribution. Residuals (not evaluated by MAX UnMix) are negligible (Fig. 1a).

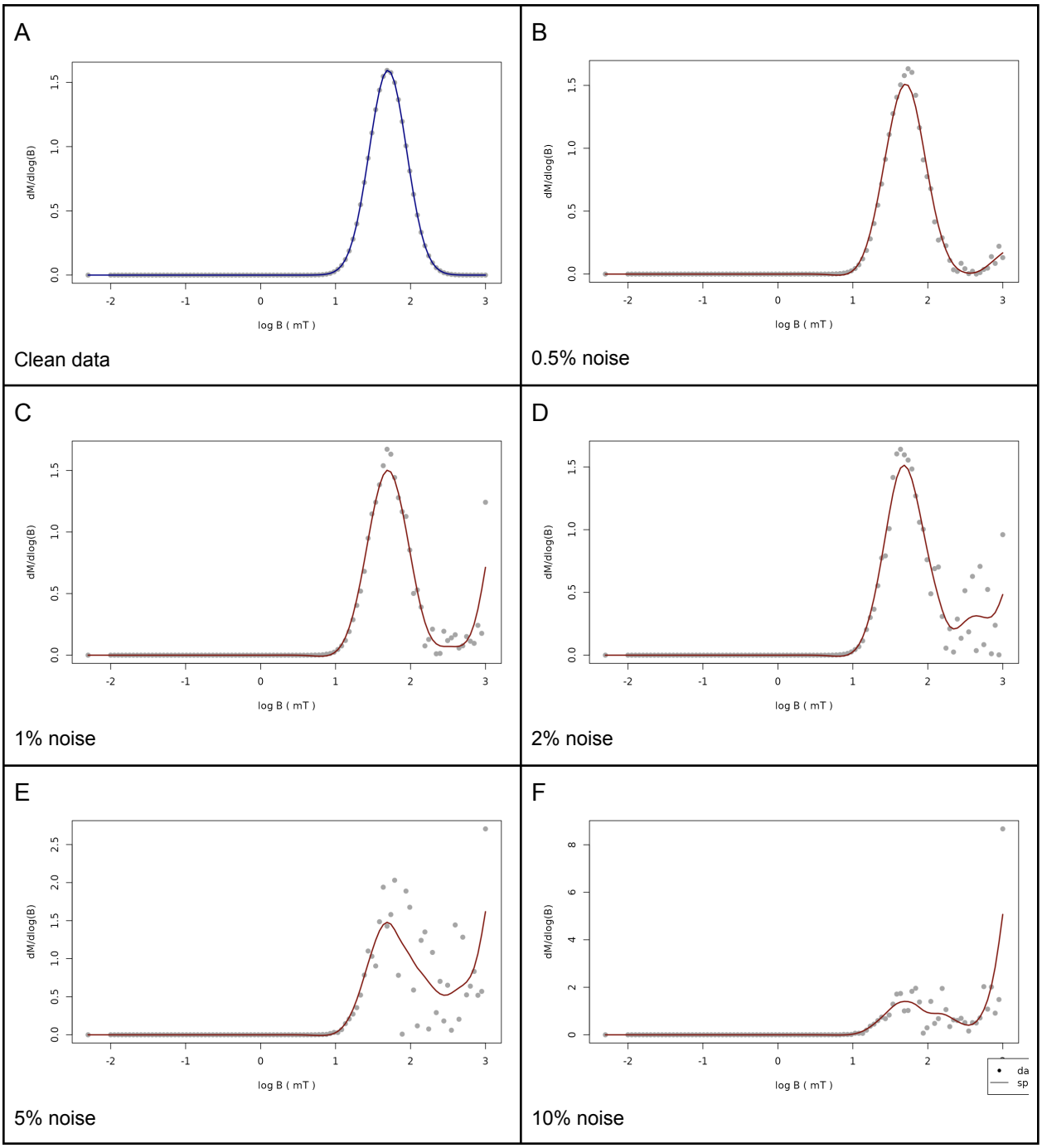


Figure 1. Synthetic single-component coercivity spectrum derivative in the absence of noise (A), and with 0.5% noise (B), 1% noise (C), 2% noise (D), 5% noise (E), and 10% noise (F) generated in MAX UnMix.

Increasing noise

The behavior changes rapidly once noise is introduced, as observed in figures 1 b-f. Two systematic effects appear: 1) the high-coercivity tail of the distribution becomes progressively non-zero, and 2) at higher noise levels, the tail begins to resemble a secondary component, yet

the synthetic dataset contains only one magnetic population. The apparent additional component arises from the interaction of measurement noise, derivative amplification, and the absolute-value transformation. But there is more.

5. Spline smoothing and edge effects

To mitigate derivative noise, MAX UnMix allows users to apply spline smoothing using the R function smooth.spline(), as shown by the red curves in figures 1b-e above. The smoothing factor *sf* can vary between 0 and 1, where *sf* = 0 is described as equivalent to no smoothing. Smoothing factors of 0.5 were used in all examples other than the clean data.

However, spline representations have a well-known limitation: their behavior near dataset boundaries is poorly constrained. Because edge points lack neighbors on one side, spline curvature can become unstable at the first or last data point. In the synthetic experiments shown here this behavior can be observed as the last point diverging upward, but in other datasets it may be the first point to become unstable, as personally observed at times. Which edge becomes unstable depends largely on the particular realization of noise and is sometimes dependent on whether IRM acquisition curves or backfield curves are measured, each with different expressions of measurement noise.

6. Even “zero smoothing” may still alter derivatives

Interestingly, the instability persists even when smoothing is set to *sf* = 0, suggesting that the derivative may still be evaluated from a spline representation of the data, rather than directly from the raw measurements. If this is the case, the derivative being analyzed is not strictly

$$\frac{dM}{d\log B} \tag{5}$$

but rather

$$\frac{d}{d\log B} f(B) \tag{6}$$

where *f*(B) is the spline approximation.

Because spline functions are inherently unstable at boundaries, the derivative at the first or last data point can become strongly biased. The latter is the case here, as it will be demonstrated below.

7. Verifying the data points

The synthetic datasets used in this experiment contain exactly 100 acquisition points. Inspection of the MAX UnMix output suggests that the derivative is evaluated at the original field values, implying that the number of points is preserved. This behavior is consistent with a spline-based derivative rather than a simple finite-difference scheme, and has important consequences for the stability of the derivative near dataset boundaries. In effect, the final point in the dataset becomes particularly sensitive to noise, as the derivative near dataset boundaries is poorly constrained. This instability can amplify small fluctuations, which are then transformed into positive signal by the absolute-value operation.

8. A contrasting workflow

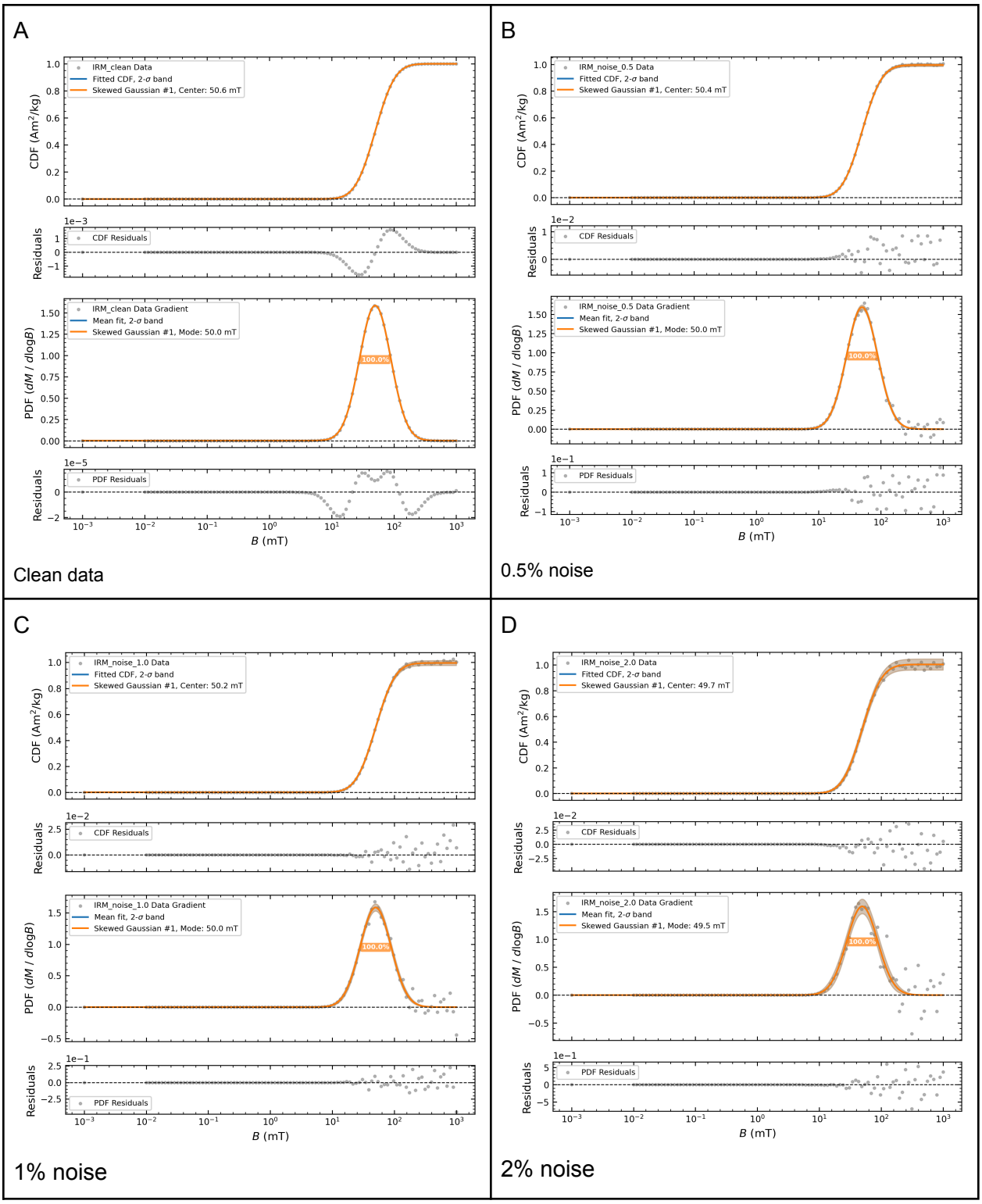
The same datasets were analyzed using CoercivityUnmixing (Bilardello, 2025), a Python-based workflow designed to fit both the cumulative acquisition curve and its derivative simultaneously without applying any smoothing of the data.

Unlike MAX UnMix, which fits only the derivative representation, CoercivityUnmixing explicitly models both domains: the CDF (IRM acquisition curve) and the PDF (coercivity spectrum).

The optimization iterates between the two representations until both the cumulative curve and its derivative are reproduced consistently. Several methodological differences follow from this approach: derivatives are not forced to be positive; both PDF and CDF domains are fitted simultaneously; model parameters must satisfy the constraints imposed by both representations.

Because the acquisition curve must also be reproduced by the model, it acts as a strong constraint on the allowable coercivity distributions. Components that might improve the fit to noise in the derivative are rejected if they degrade the fit to the measured acquisition curve.

The workflow additionally also plots residuals for both domains. For each same dataset used above, Figure 2 displays both the fitted CDF and PDF and respective residuals obtained from the same initial component-fitting guesses, providing an immediate diagnostic of whether the model is fitting signal or noise.


A
IRM_clean Data
Fitted CDF, 2-σ band
Skewed Gaussian #1, Center: 50.6 mT
CDF (Am²/kg)
Residuals
CDF Residuals
IRM_clean Data Gradient
Mean fit, 2-σ band
Skewed Gaussian #1, Mode: 50.0 mT
100.0%
PDF (dM / dlogB)
PDF Residuals
B (mT)
Clean data
B
IRM_noise_0.5 Data
Fitted CDF, 2-σ band
Skewed Gaussian #1, Center: 50.4 mT
IRM_noise_0.5 Data Gradient
Mean fit, 2-σ band
Skewed Gaussian #1, Mode: 50.0 mT
0.5% noise
C
IRM_noise_1.0 Data
Fitted CDF, 2-σ band
Skewed Gaussian #1, Center: 50.2 mT
IRM_noise_1.0 Data Gradient
Mean fit, 2-σ band
Skewed Gaussian #1, Mode: 50.0 mT
1% noise
D
IRM_noise_2.0 Data
Fitted CDF, 2-σ band
Skewed Gaussian #1, Center: 49.7 mT
IRM_noise_2.0 Data Gradient
Mean fit, 2-σ band
Skewed Gaussian #1, Mode: 49.5 mT
2% noise

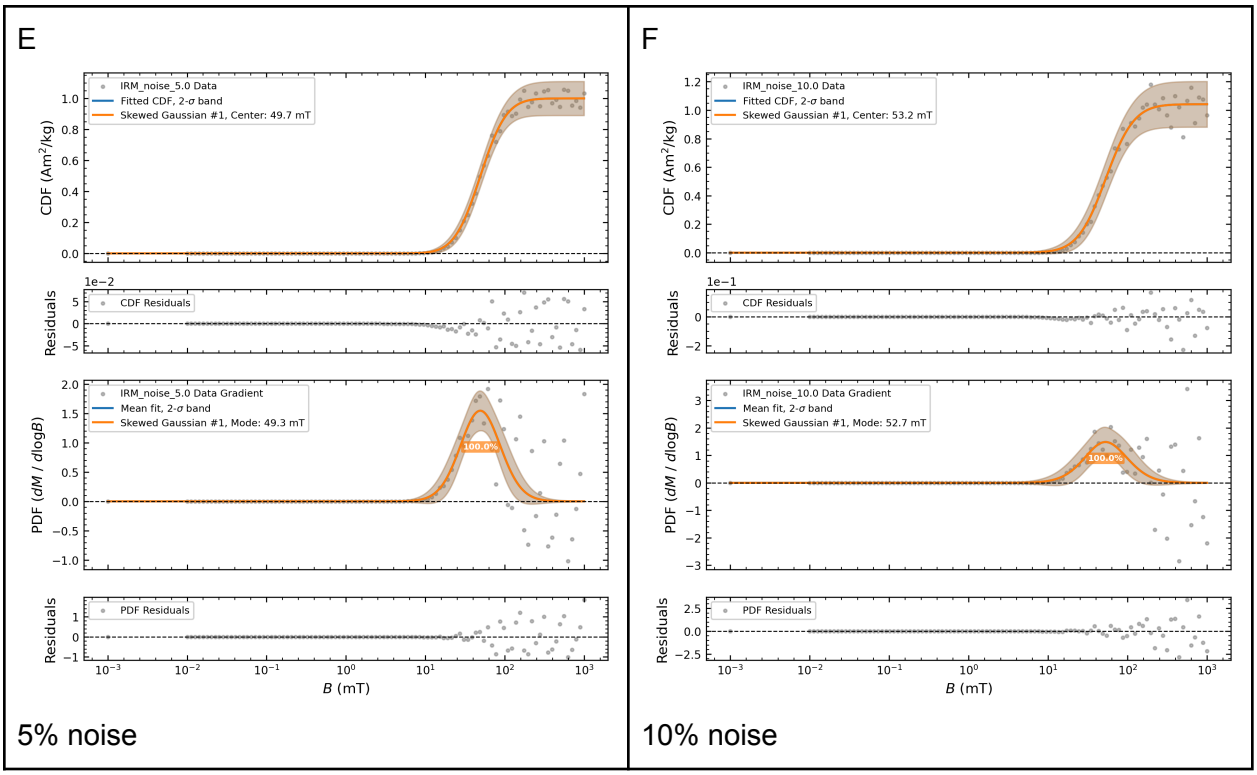


Figure 2. Same synthetic single-component coercivity spectrum as in Fig. 1 fitted and plotted using CoercivityUnmixing: CDF (data), PDF (gradient) and relative residuals in the absence of noise (A), and with 0.5% noise (B), 1% noise (C), 2% noise (D), 5% noise (E), and 10% noise (F). The same initial component-fitting guesses were used to generate each panel.

In the synthetic datasets analyzed here, the results show that: the cumulative acquisition curves are reproduced accurately across all noise levels; the derivative spectra retain symmetric noise around zero in the tails; residuals remain centered on zero rather than systematically positive.

Even at higher noise levels, the derivative does not develop artificial positive tails. Instead, the fluctuations remain statistically symmetric around zero, reflecting the noise properties of the original data. This behavior contrasts sharply with the MAX UnMix results, where the absolute derivative systematically transforms derivative noise into positive signal.

Moreover, it can be observed that the scatter in the data does not increase as dramatically for the last points here as they do with MAX UnMix, whether smoothing is expressly applied or not, confirming that even selecting an *sf*= 0 alters the distribution of the gradient data.

The simultaneous fitting of both domains therefore provides a simple but powerful safeguard: a proposed coercivity distribution must not only reproduce the derivative spectrum, but must also remain consistent with the measured acquisition curve and its residual structure. Because both domains are evaluated simultaneously, residual patterns that might otherwise go unnoticed in the derivative alone become immediately visible in the cumulative curve and its residuals.

9. Peer review and methodological assumptions

The definition of the coercivity spectrum used in MAX UnMix is stated explicitly in the article, which means that the positivity assumption was present in the manuscript from the beginning.

The paper underwent peer review by experienced researchers in rock magnetism, and it is reasonable to assume that the reviewers examined the statistical formulation carefully. However, peer review is not infallible. Even thorough reviewers occasionally overlook subtle methodological implications, particularly when an assumption follows established plotting conventions. It is also possible (and unfortunately too common) that not all reviewer comments were fully reflected in the final version of the article. Scientific literature contains many examples where widely used methods later required refinement once their statistical properties were examined more closely.

10. Why openness matters

Analytical software is not just a convenience; it is part of the scientific method. When algorithms are implemented in transparent, inspectable code, methodological assumptions can be evaluated and improved by the community. However, when tools are distributed primarily as web applications, that process becomes more difficult.

The rapid adoption of MAX UnMix, as funding to other online processing tools, demonstrates the demand for accessible analytical tools. It also illustrates the importance of ensuring that such tools remain open to ongoing scrutiny and revision.

11. Revisiting coercivity spectra: how many coercivity components are actually noise?

The synthetic experiment presented here contains only a single magnetic component. Yet once modest levels of measurement noise are introduced, the derivative tails generated by the MAX UnMix workflow begin to resemble additional coercivity populations. At sufficiently high noise levels the distribution can appear to contain two or even more components.

Nothing in the underlying “physics” (if that can be said of synthetic data) has changed. The additional structure arises entirely from the interaction of derivative amplification of noise, spline boundary behavior, and the absolute-value transformation applied to the derivative. The result is that noise in the tails of the distribution is systematically converted into positive signal.

This raises a simple but uncomfortable question: how many published coercivity components fitted using MAX UnMix are actually noise?

The derivative-tail bias described here is most significant in datasets with weak magnetizations, including sediments and sedimentary rocks, speleothems and other environmental magnetic archives, as well as iron speciation studies to name a few examples. In these materials, the signal in the derivative tails is often comparable to measurement noise. Under such conditions, the positive bias introduced by the absolute derivative can produce artificial features that resemble additional coercivity components.

Coercivity unmixing has been widely applied to precisely these types of materials. This does not imply that published interpretations are necessarily incorrect, as studies should rely on multiple lines of rock-magnetic evidence. However, the possibility remains that some weak coercivity components, particularly those appearing in distribution tails, may reflect statistical artifacts rather than mineral physics.

Re-examining such datasets with derivative methods that preserve noise symmetry may therefore lead to simpler and more physically realistic interpretations. Considering the widespread use of MAX UnMix over the past decade, such a re-evaluation becomes increasingly important if not a necessity.

12. The derivative alone is not enough

Another implication follows from the behavior of derivatives themselves. The derivative representation is attractive because it visually separates overlapping components. Peaks that are difficult to distinguish in the cumulative acquisition curve can appear as distinct distributions in the gradient plot. However, numerical derivatives inherently amplify noise so that when measurement noise is present, fitting the derivative alone can therefore become something of a guessing game. Because the derivative emphasizes small fluctuations, it is always possible to introduce additional components that improve the local fit to noisy peaks.

The cumulative acquisition curve provides an essential complementary constraint. Unlike its derivative, the acquisition curve represents the original measured magnetization and therefore retains the statistical properties of the measurements themselves. If additional components are introduced in the derivative domain merely to match noise, they must still reproduce the measured acquisition curve.

When such components are introduced, the resulting model often produces unnecessary curvature or oscillations in the cumulative fit, indicating that the model is fitting noise rather than signal. Very often, fitting unnecessary components in the PDF results in an integrated (not fitted) CDF that overshoots the true magnetization of the data. When fitting the PDF only as in MAX Unmix, it is particularly common to “be forced” to fit low coercivity components that have low amplitude and large dispersion (large sigma) in order to fully capture the initial curvature of the gradient data. Such components would be erroneously interpreted as either highly viscous low-coercivity multi-domain particles and/or viscous superparamagnetic grains. Conversely, fitting components to the tail-end of the distribution, which typically and diagnostically never

goes down to zero as observed here, naturally leads to the interpretation of unsaturated high-coercivity mineral components (hematite or goethite).

Thus the acquisition curve provides a straightforward diagnostic test: If the cumulative fit remains smooth and consistent with the measured data, the component structure is likely justified; If the cumulative fit begins to show unnecessary complexity and has larger magnetization than the underlying data, the model is overfitting derivative noise.

In practice, the derivative spectrum should therefore be viewed primarily as a diagnostic visualization when fitting the data, while the physical validity of the model must ultimately be judged against the measured acquisition curve.

13. Science corrects itself

Scientific tools evolve, and widely used methods are occasionally refined as their statistical properties become better understood. The rapid adoption of coercivity unmixing techniques demonstrates how valuable these approaches are to the rock-magnetic community.

At the same time, the present analysis illustrates how subtle algorithmic choices, such as taking the absolute value of a derivative or relying exclusively on gradient fitting, can influence the interpretation of coercivity spectra. The solution is straightforward: transparent algorithms; reproducible workflows; open implementations that allow the community to test and refine analytical methods.

Sometimes the most interesting discoveries are not in the peaks of a coercivity spectrum, but in the behavior of its tails.

References and software

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All data used in this article and corresponding CoercivityUnmixing fits can be found at:
https://github.com/DBilardello/Rock-Paleo-Magnetism/tree/main/RPM%20Newsletter

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